

Facial Emotion Recognition

<https://github.com/aiyshwary/Facial-Emotion-Detection>

Python

CNN

SVM

Random Forest

TensorFlow

Jupyter

OVERVIEW

A comparative study of machine learning and deep learning approaches for recognizing facial emotions from grayscale images, benchmarking CNN, SVM, Random Forest, and Decision Tree models across multiple datasets.

IMPACT

Implemented and benchmarked four ML/DL models for 7-class facial emotion classification. CNN achieved the best generalization at approximately 75.5% accuracy, outperforming all traditional models on unseen data.

KEY DETAILS

- Trained on icml_face_data and Kaggle FER datasets with 7 emotion classes (happy, sad, angry, fear, surprise, disgust, neutral).

- CNN with data augmentation and stratified cross-validation achieved approximately 75.5% validation accuracy.

- Random Forest reduced overfitting vs Decision Tree, achieving approximately 89.2% on ICML dataset.

- SVM with PCA preprocessing achieved approximately 55% accuracy — lowest among all models on high-dimensional pixel data.

- Concluded CNNs are best suited for FER due to hierarchical spatial feature extraction.